Communicating with Data – Storytelling

Dr. Ab Mosca (they/them)

Slides based off slides courtesy of Jordan Crouser (https://jcrouser.github.io/)

Plan for Tuesday/ Wednesday

- Ab is traveling for a conference
- **No office hours Tuesday** (you can reach out on Slack if you need anything emergently)
  - Professor Crouser has office hours in Bass 105 Thurs. 12:30-2:00, and Fri. 11:00-12:00 he can answer 109 questions if you stop by
- **No class Wednesday** but you will have a Tableau tutorial to complete during class time

Plan for Today

- Narrative structures
- Narrative structures in visualization
- Roles of the reader and the author

Storytelling with ~~visualization~~



Discussion

storyteller



What makes a **good story**?



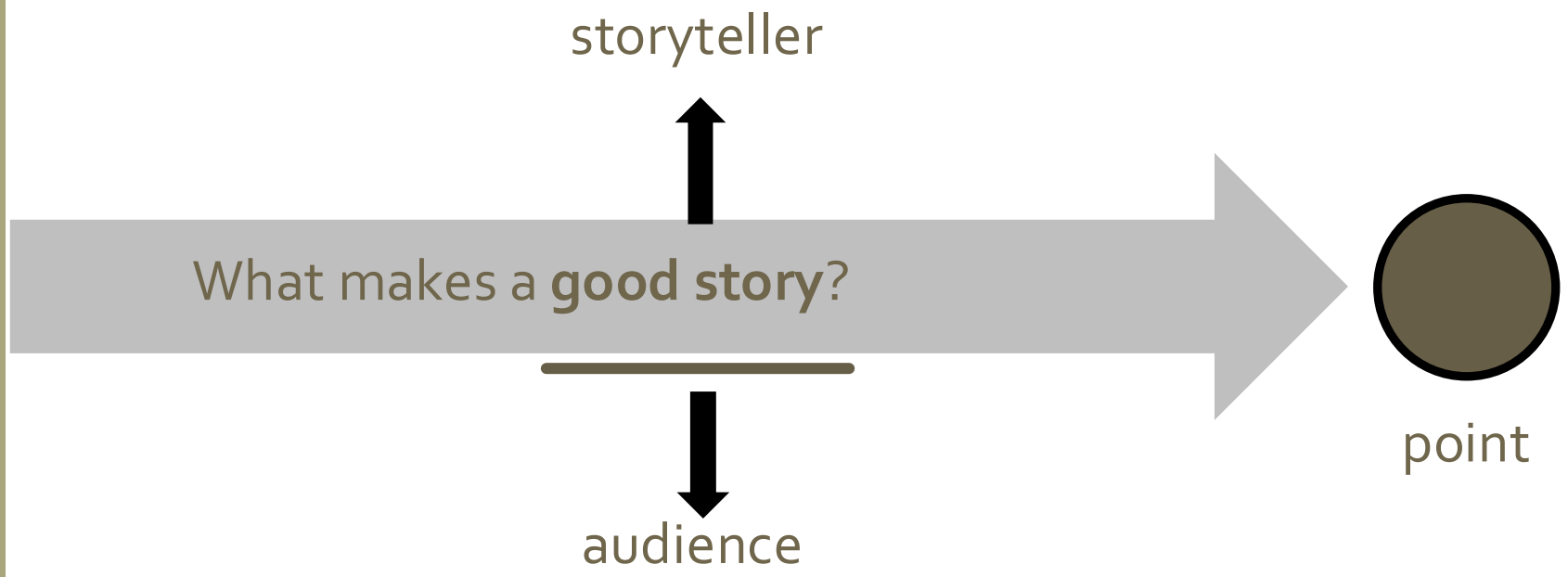
audience

Story teller & Audience

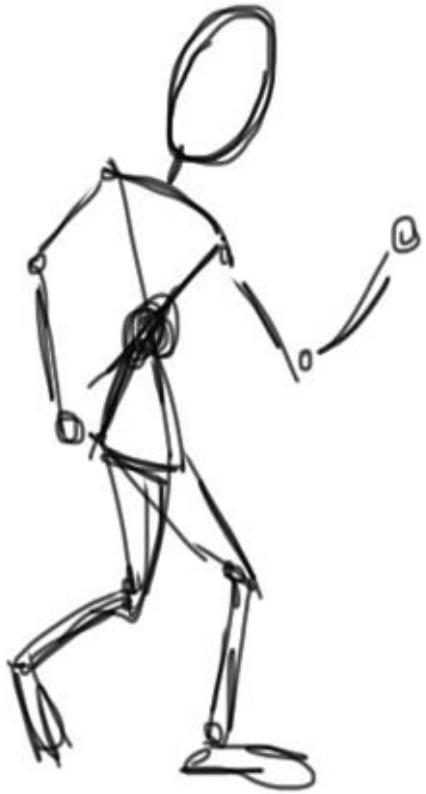




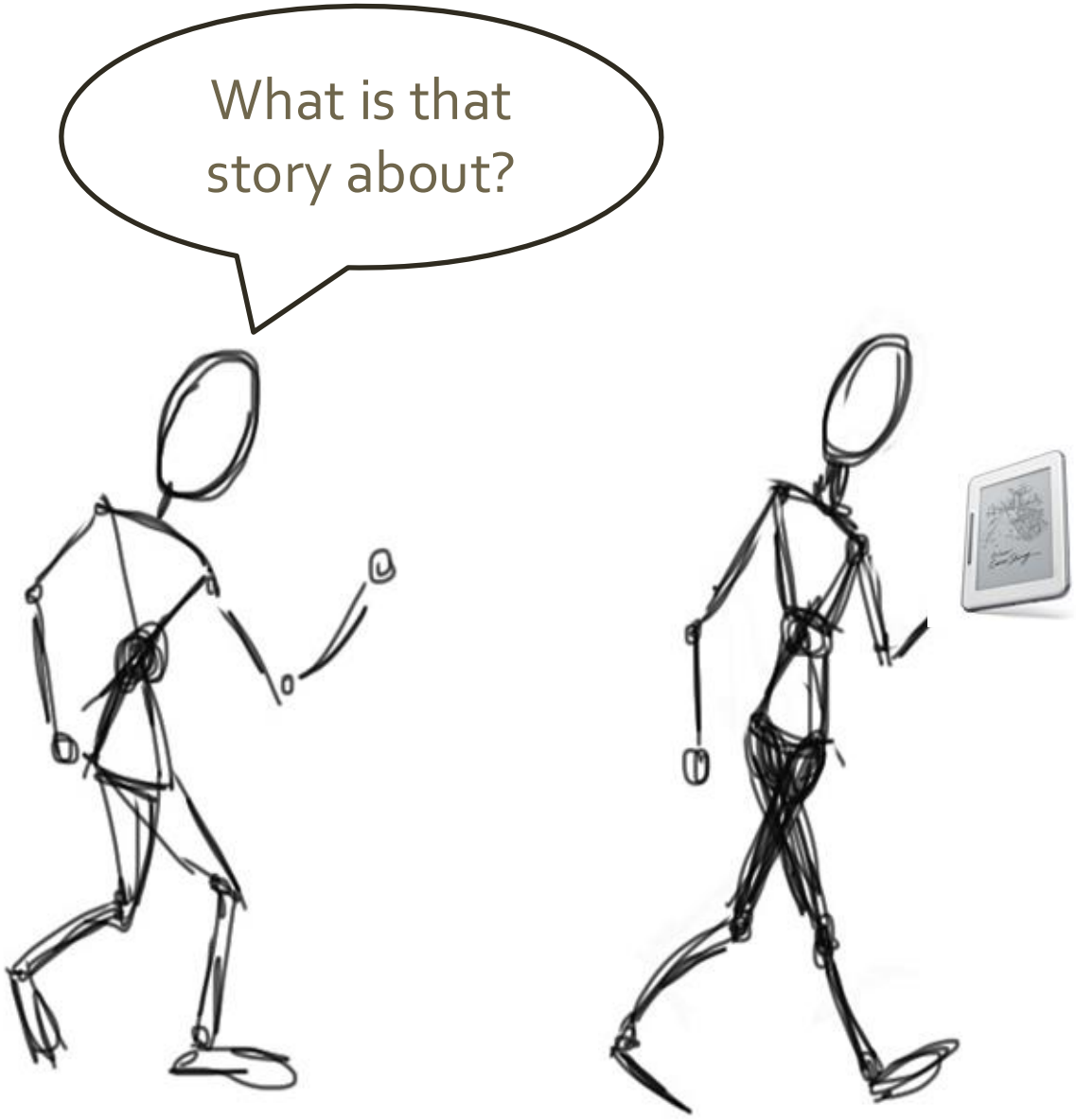
Discussion

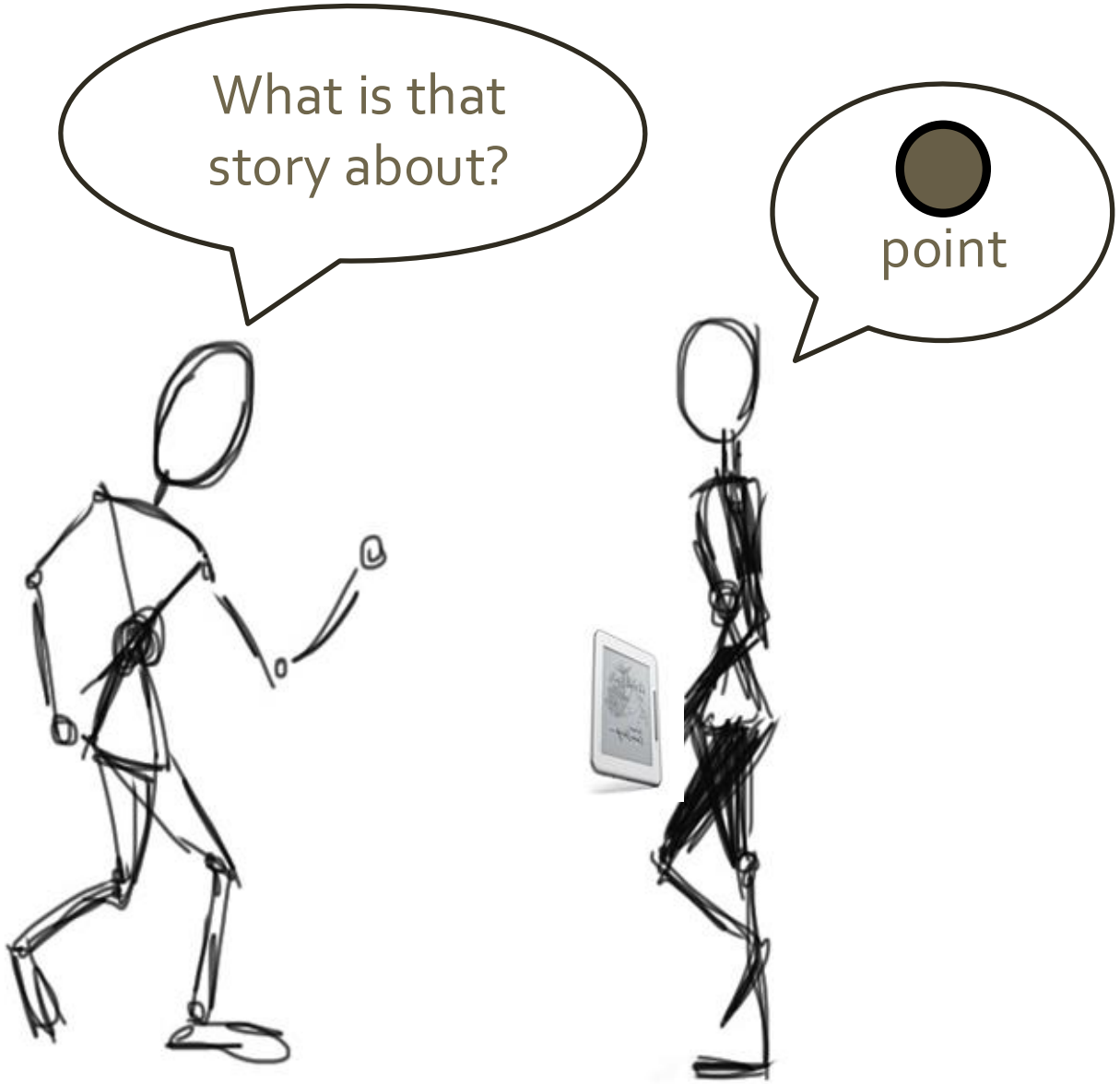


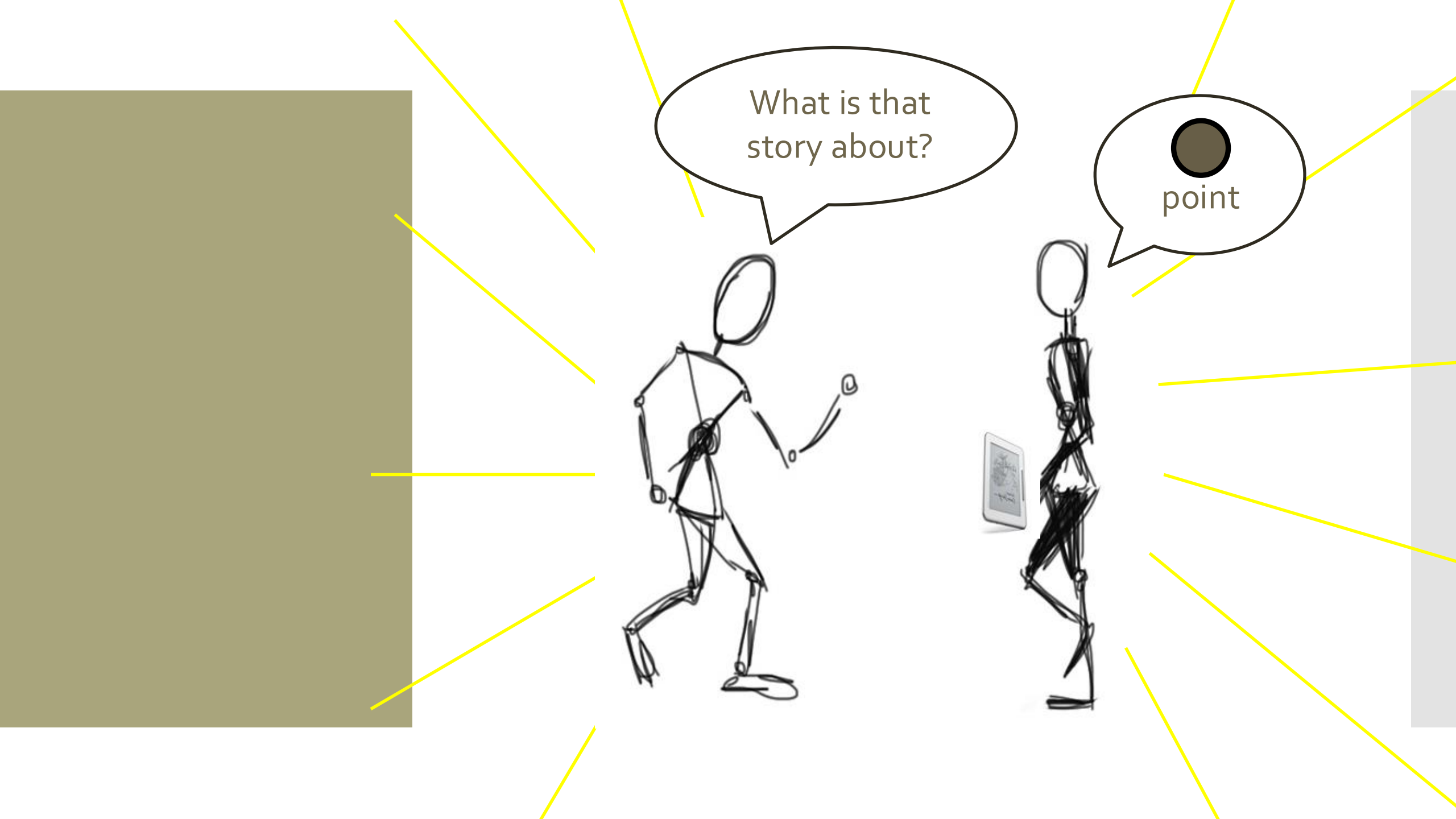




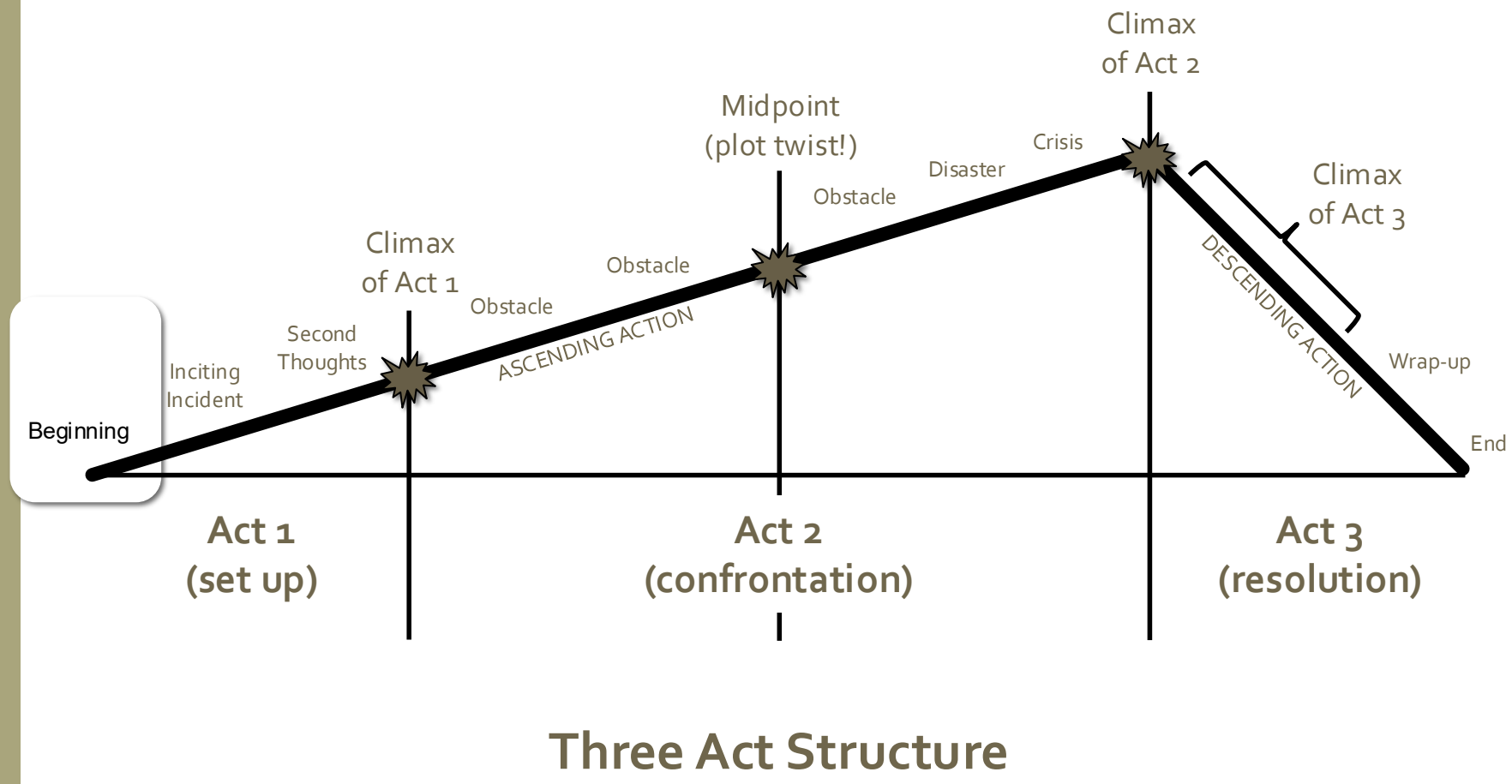



What is that story about?


What is that story about?
point

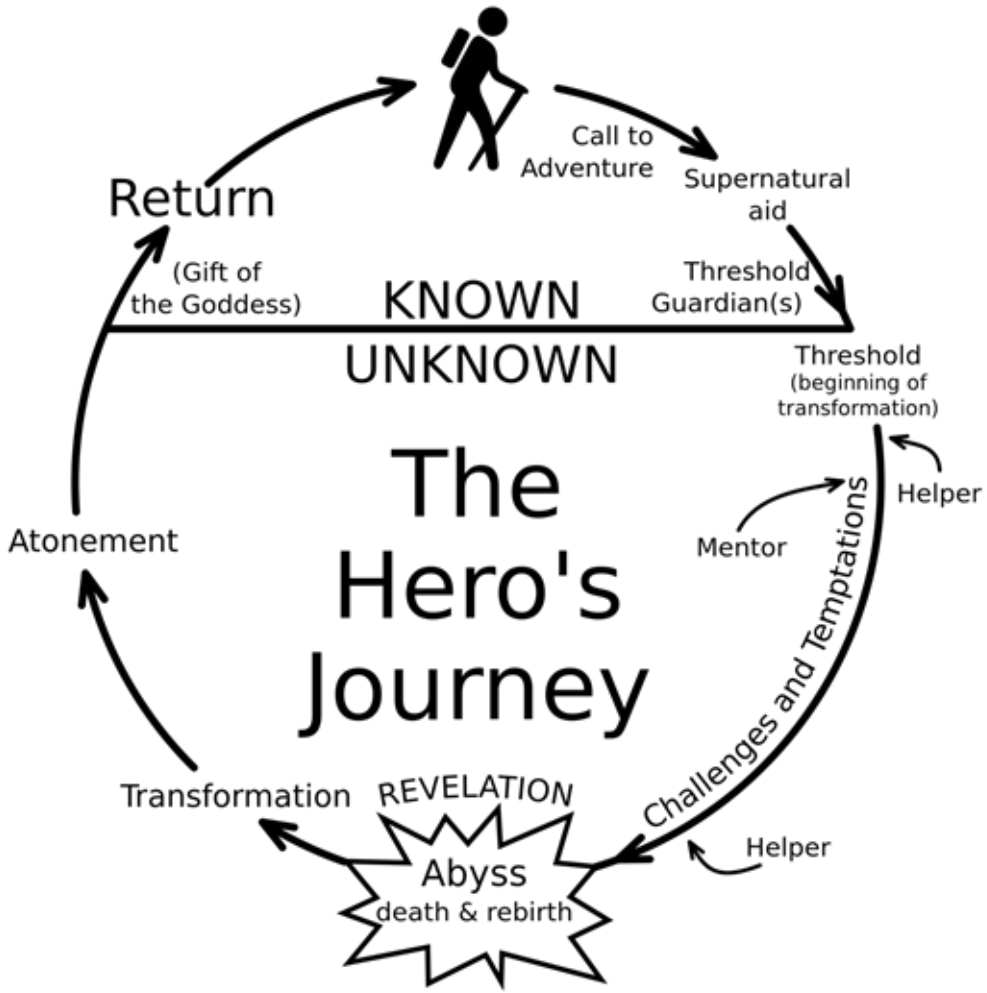

What is that story about?
point

Ways to get to the point

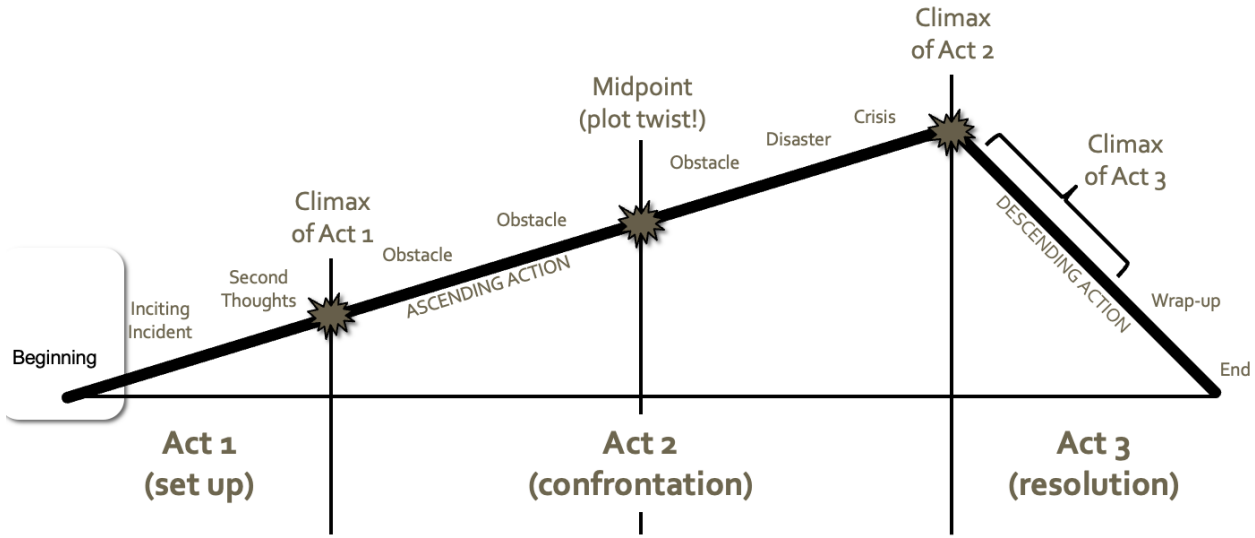


Three Act Structure

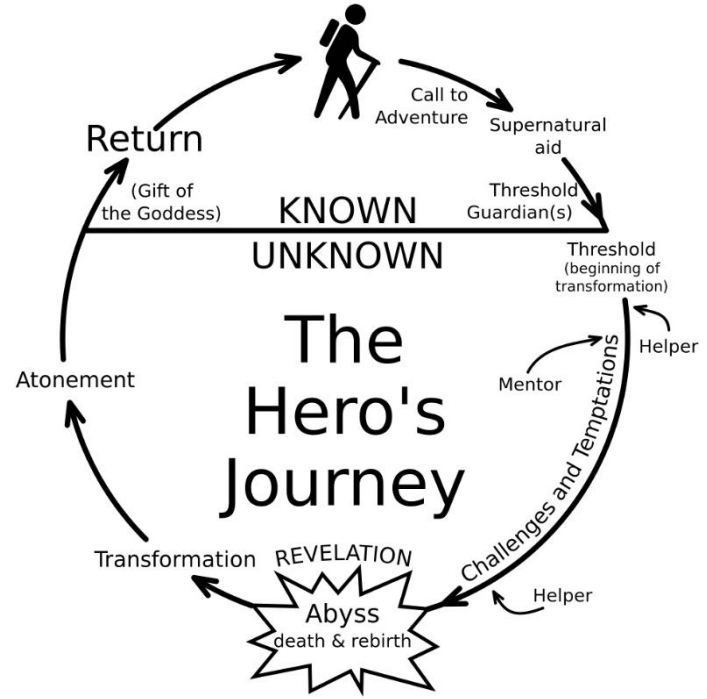
Ways to get to the point



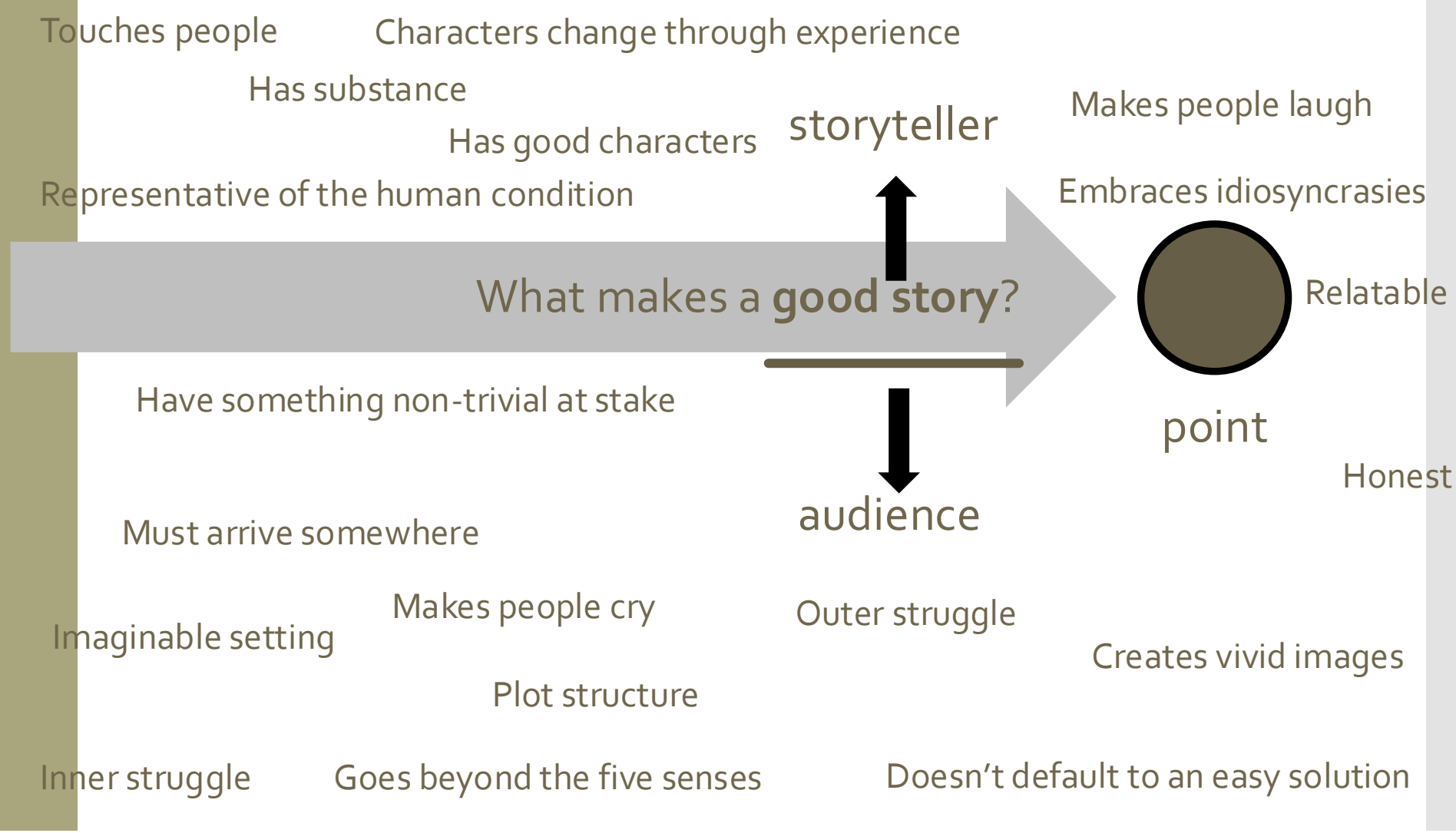
What movies use what arc?



Three Act Structure



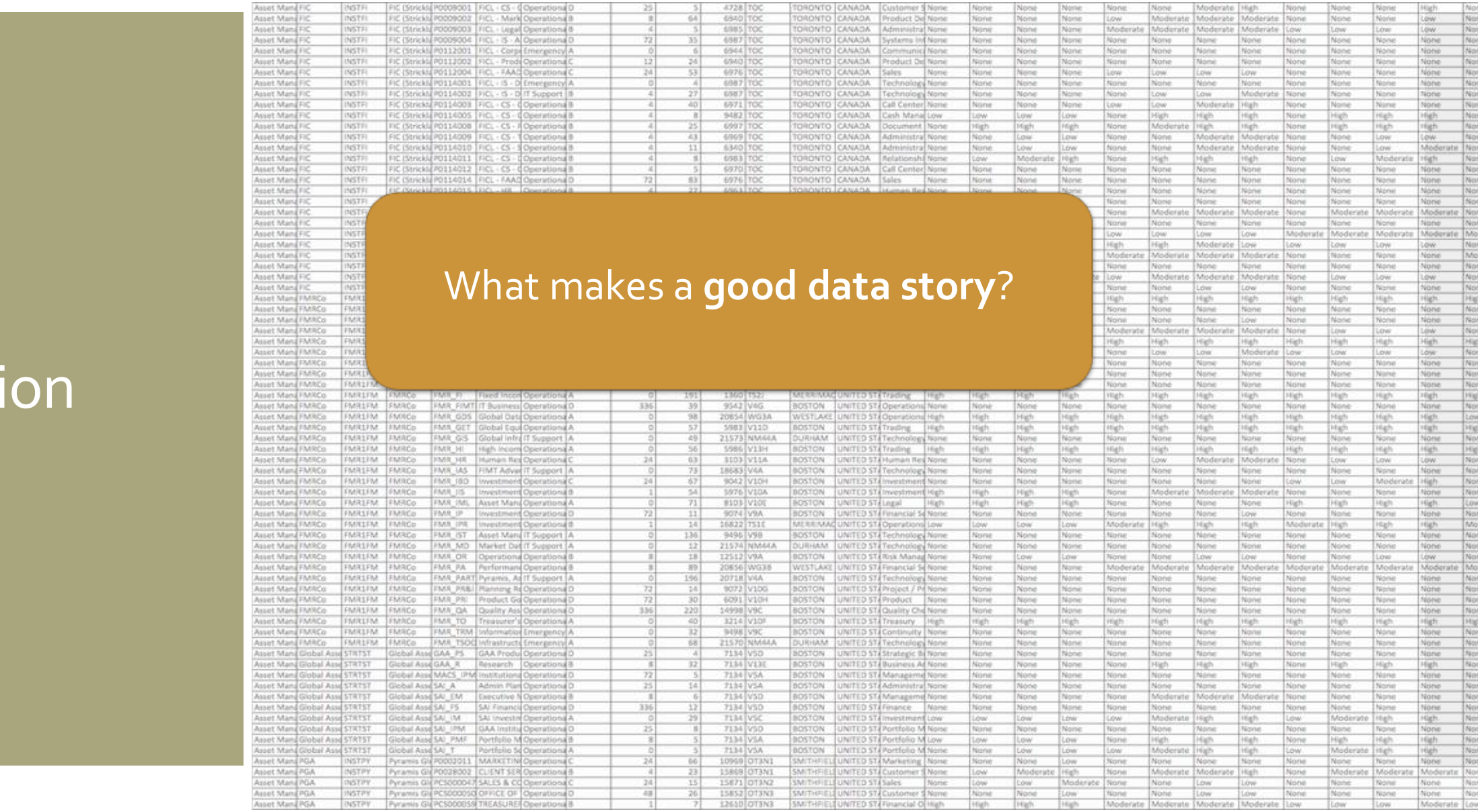
And a bunch of other stuff, too…



Storytelling with visualization

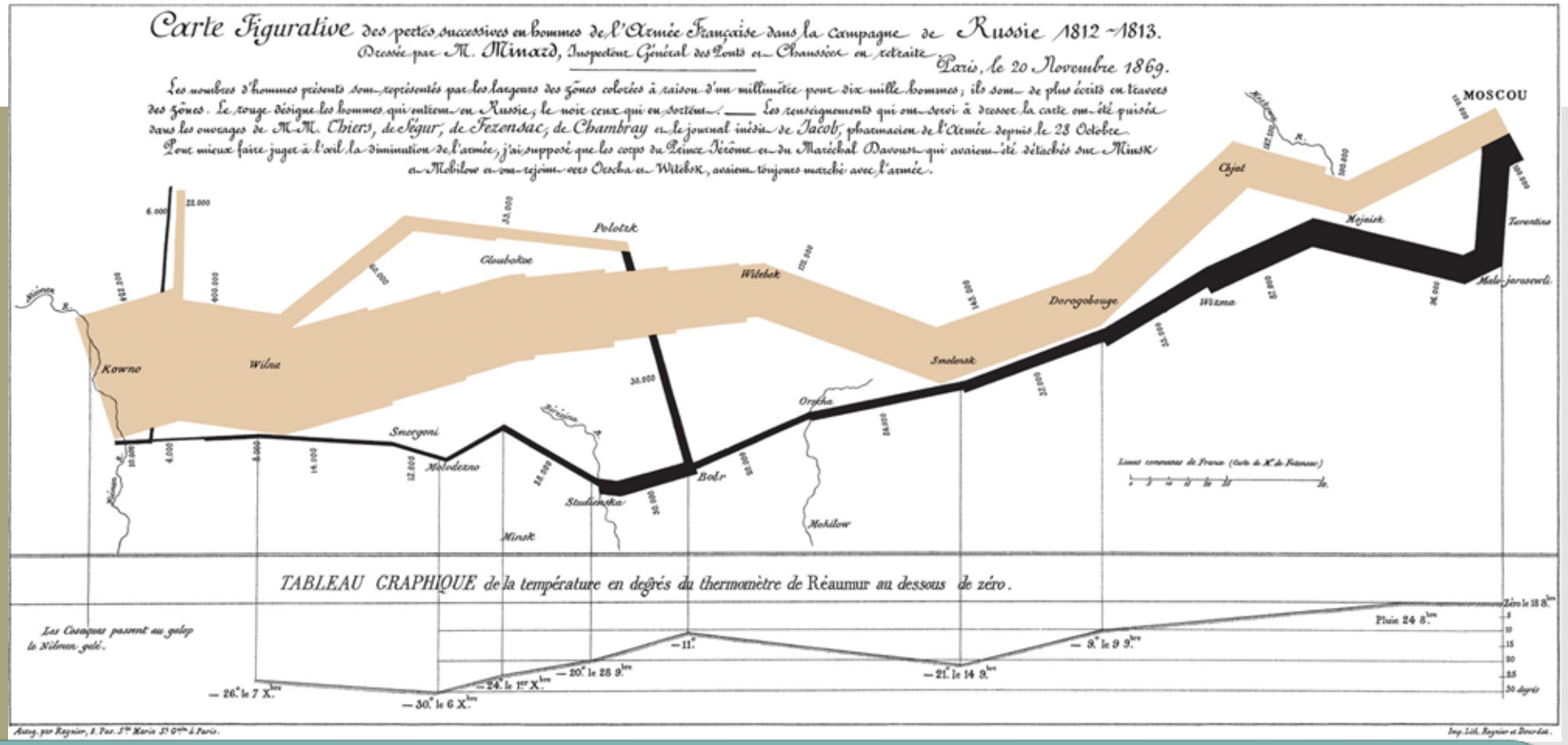


Discussion



What makes a **good data story**?

Example



"Probably the best statistical graphic ever drawn, this map by Charles Joseph Minard portrays the losses suffered by Napoleon's army in the Russian campaign of 1812. Beginning at the Polish-Russian border, the thick band shows the size of the army at each position. The path of Napoleon's retreat from Moscow in the bitterly cold winter is depicted by the dark lower band, which is tied to temperature and time scales." - *Edward Tufte*

Example



https://www.reuters.com/graphics/HEALTH-CORONAVIRUS/USA-CASUALTIES/lbpggbmrapq/

Example

SAVING THE SEQUOIAS

By **Travis Hartman**, **Ally J. Levine** and **Anurag Rao**

PUBLISHED AUG. 3, 2022

https://www.reuters.com/graphics/USA-WILDFIRES/YOSEMITE/zjvqkbqodvx/

Example

TheUpshot

Why States Are Cutting Back on Their Climate Goals

Ambitious aims have given way to political anxiety over prices as many states' 2030 emissions targets appear increasingly out of reach.

https://www.nytimes.com/2026/09/04/upshot/states-carbon-emissions-goals.html

Example

TheUpshot

Look Up Your School District's Test Scores

By Francesca Paris May 13, 2026

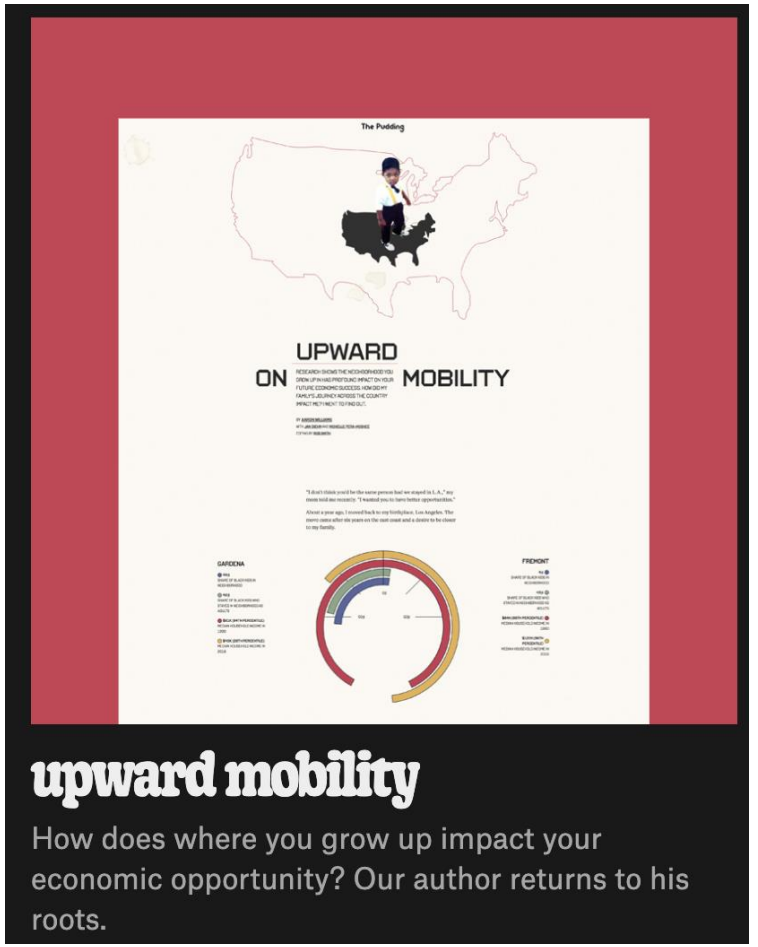
https://www.nytimes.com/interactive/2026/upshot/look-up-district-test-scores.html

Example



https://pudding.cool/2024/10/abortion-mazes/

Example

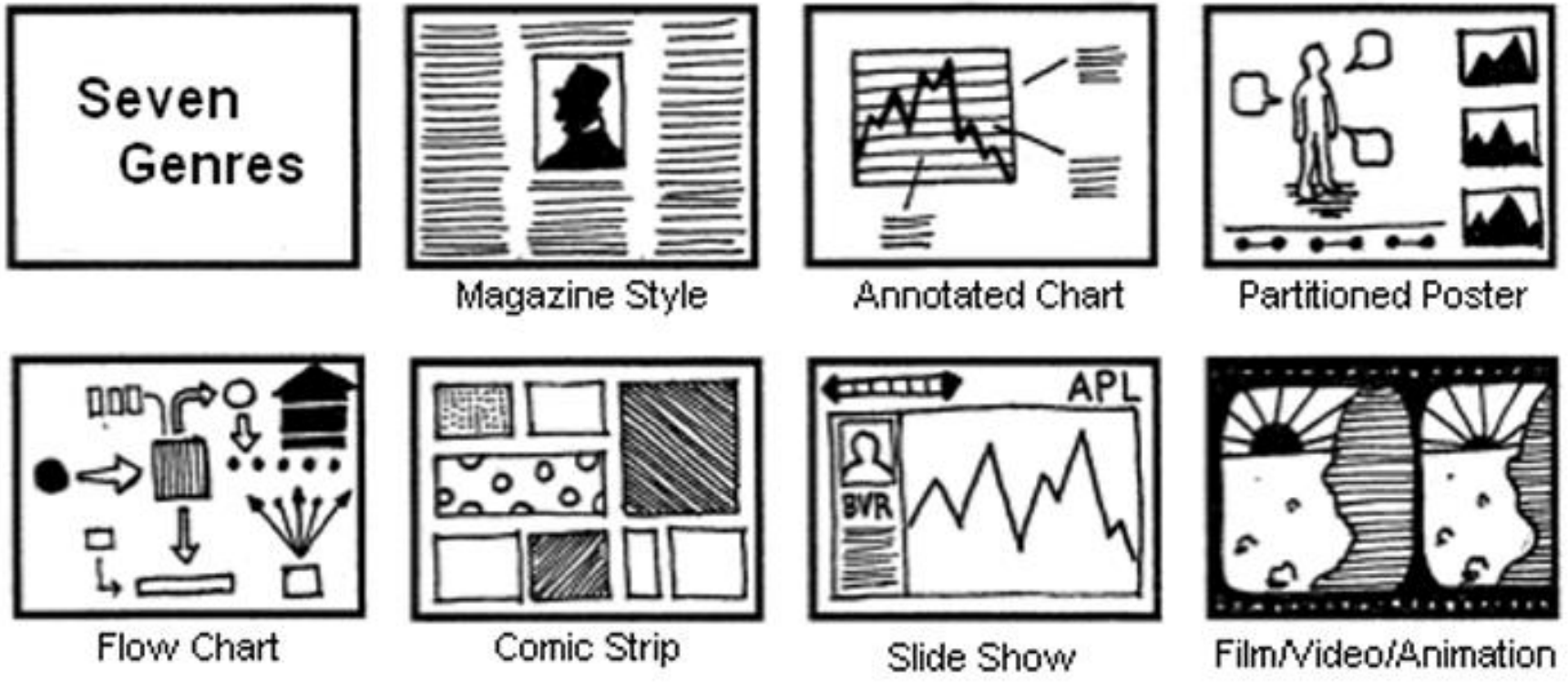


https://pudding.cool/2022/11/upward-mobility/

Design Space of Narrative Visualization

1. Genre
2. Visual Narrative Tactics
3. Narrative Structure

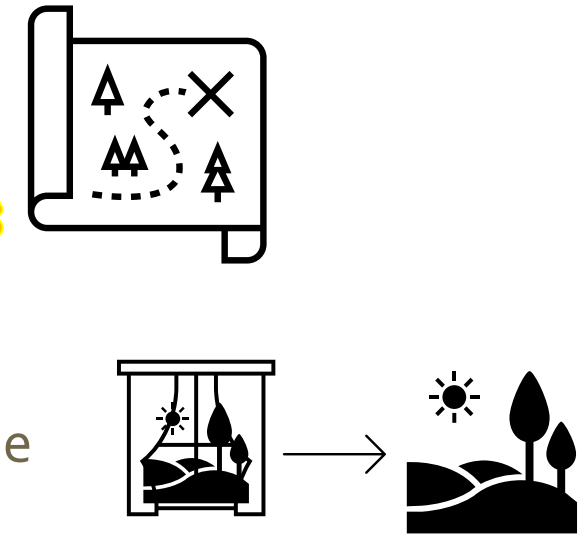
Genre



Narrative Visualization: Telling Stories with Data. Edward Segel and Jeffrey Heer

Visual Narrative Tactics

- Visual devices that assist and facilitate the narrative
  - Visual structuring
  - Highlighting
  - Transition guidance



Narrative Visualization: Telling Stories with Data. Edward Segel and Jeffrey Heer

Narrative Structure

Author Driven

Reader Driven

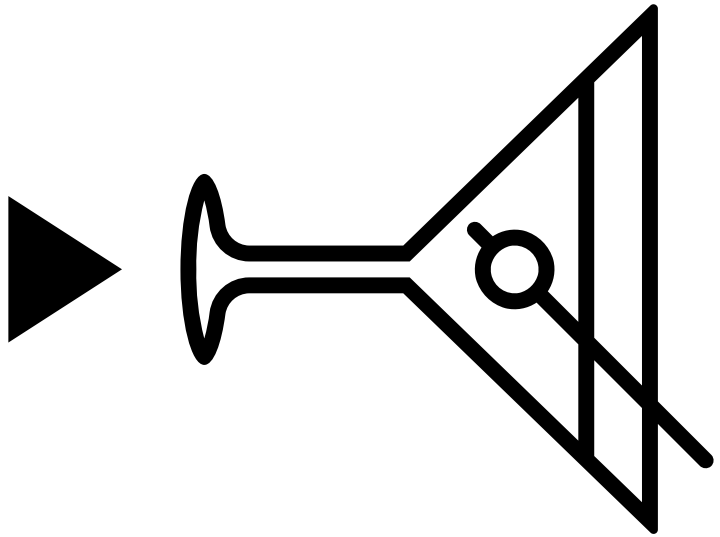


- Linear ordering of scenes
- Heavy messaging
- No interactivity

- No prescribed ordering
- No messaging
- Free interactivity

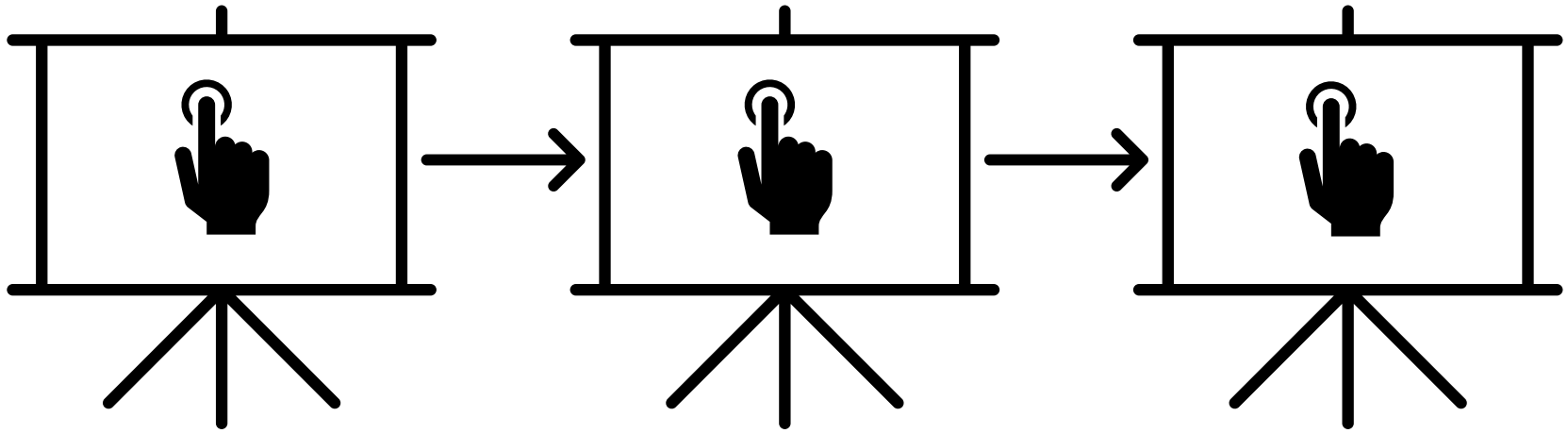
Narrative Visualization: Telling Stories with Data. Edward Segel and Jeffrey Heer

Narrative Structure: Martini Glass



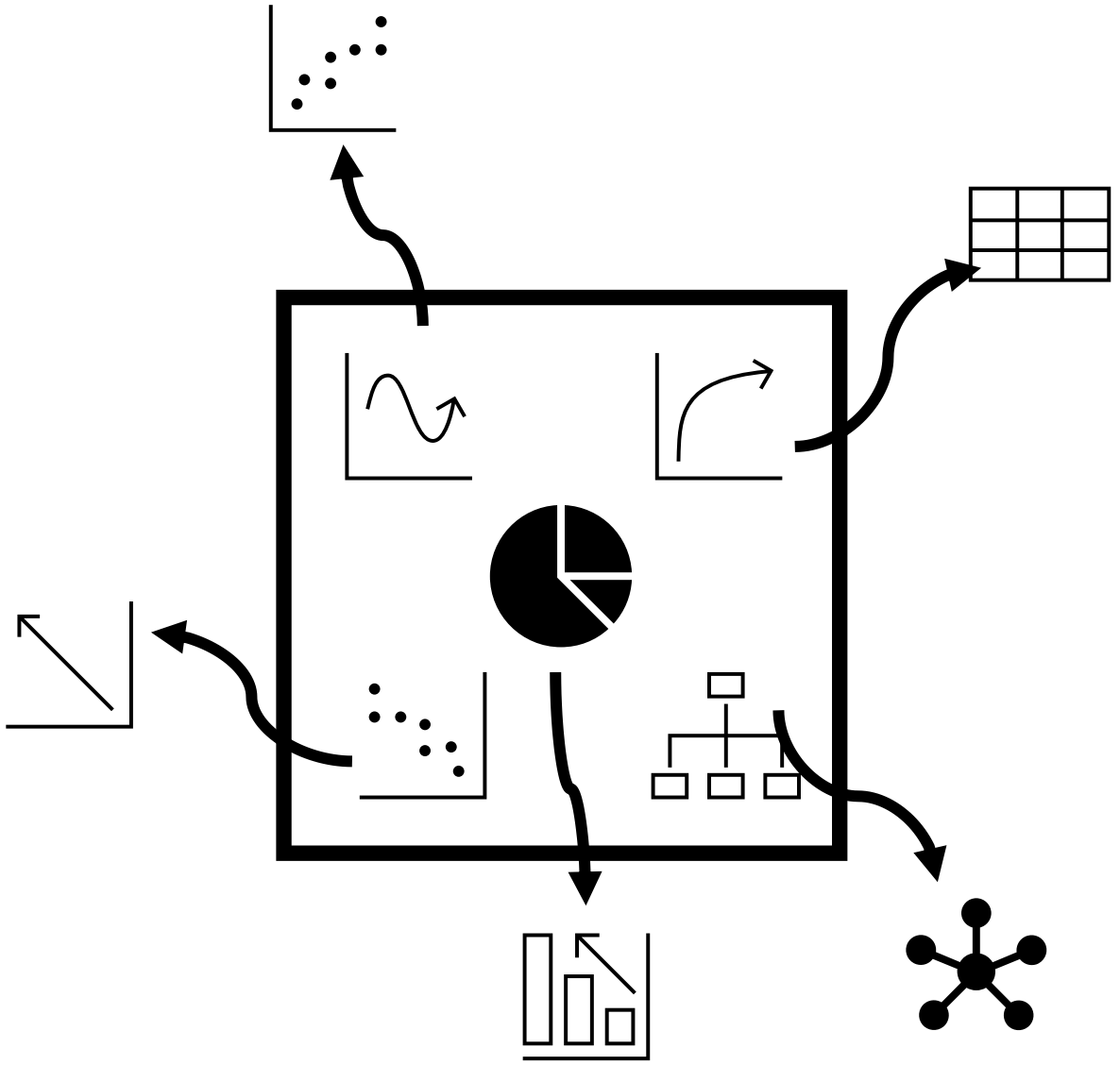
Narrative Visualization: Telling Stories with Data. Edward Segel and Jeffrey Heer

Narrative Structure: Interactive Slide Show



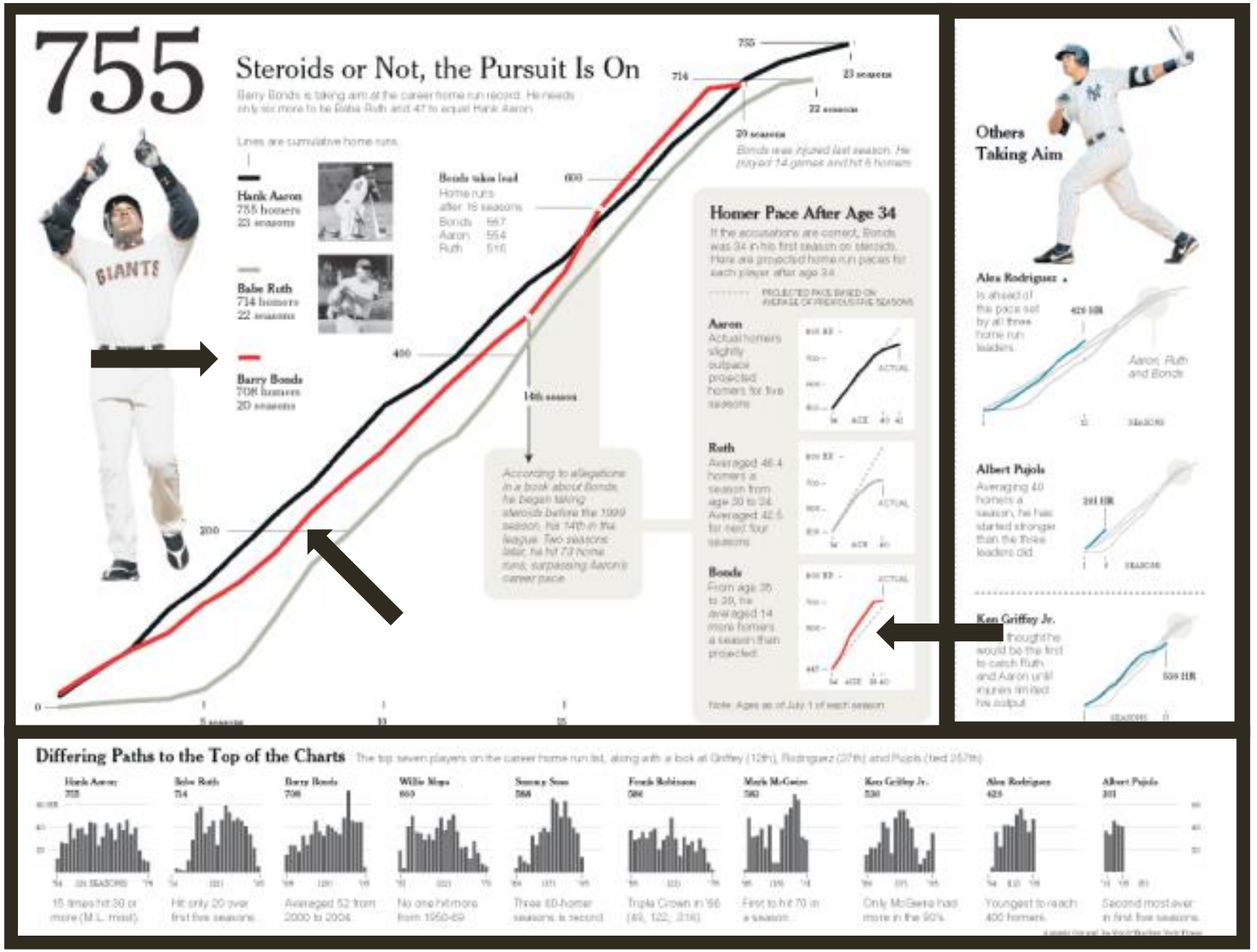
Narrative Visualization: Telling Stories with Data. Edward Segel and Jeffrey Heer

Narrative Structure: Drill-Down Story



Narrative Visualization: Telling Stories with Data. Edward Segel and Jeffrey Heer

Example: Baseball (static)

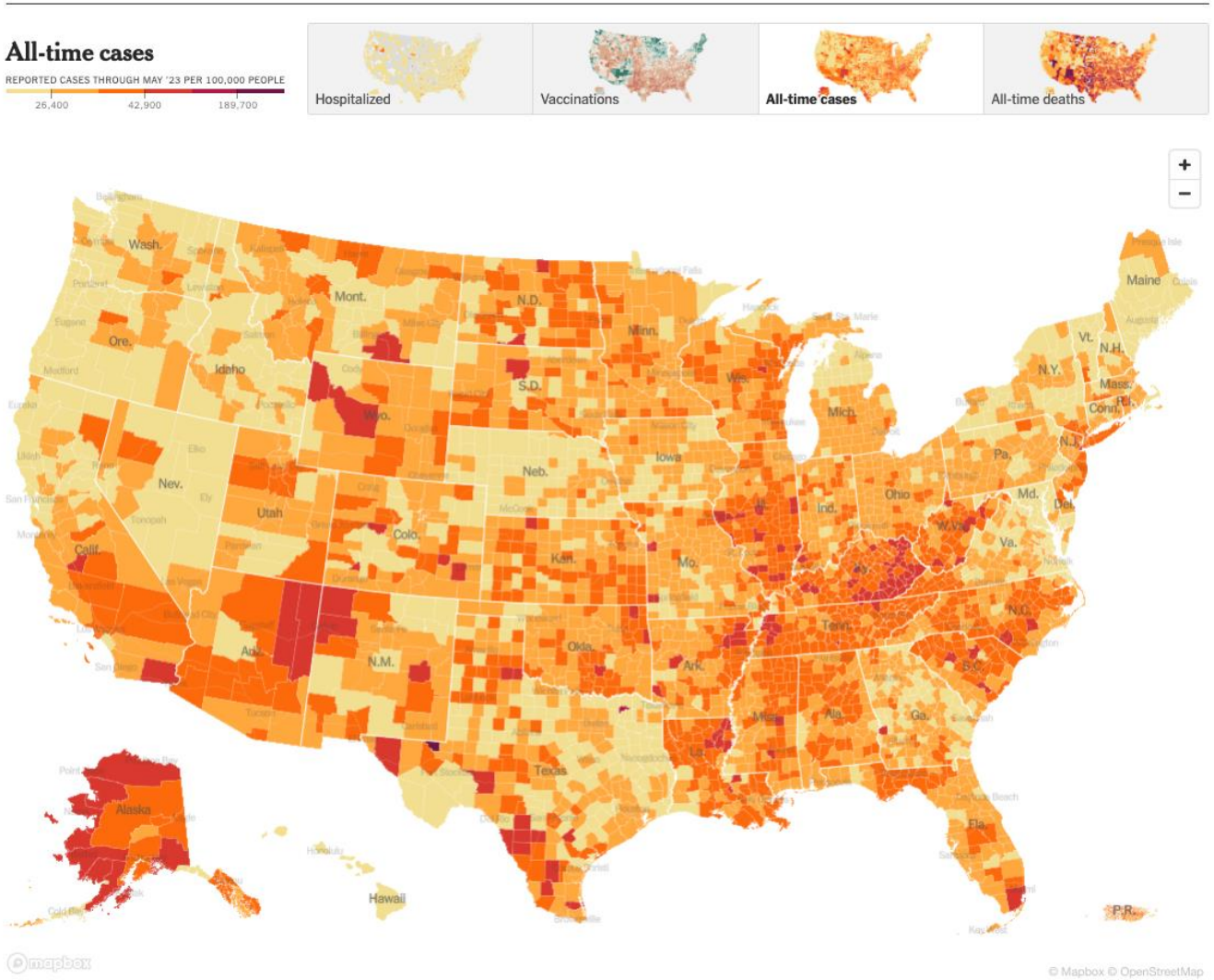


Example: Wealth inequality in the US (animated)



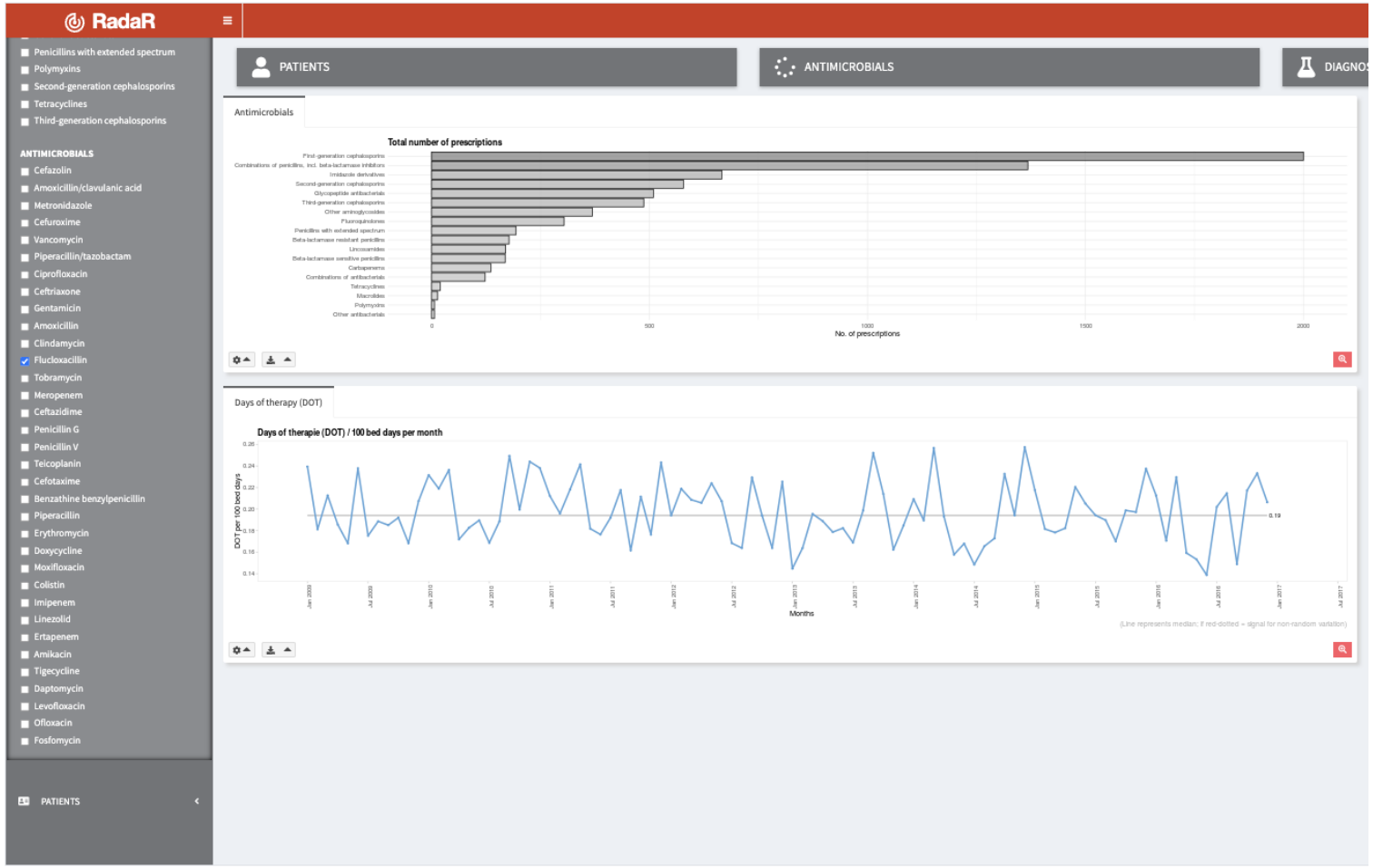
https://youtu.be/QPKKQnijnsM?si=j_hPOOsIxFGkNgHM

Example: Track Covid-19 in the US (interactive)



https://www.nytimes.com/interactive/2023/us/covid-cases.html

Example: Hospital Data (interactive)



https://shiny.posit.co/r/gallery/life-sciences/hospital-data-antimicrobial/

Practice

- Work in small groups or pairs
- Pick a topic of interest to you
- Sketch 3 possible narrative visualizations for your topic
  - One should be primary author-driven
  - One should be primarily reader-driven
  - And the last should be somewhere between